

Data Literacy

1. Water-land map with x and y scale.
 - a. Draw the probability distribution of finding gold.
 - b. If $y = -1.25$ how likely is it to find gold between $x = [-1; 0]$?
 - c. Are x and y independent?
2. Estimator
 - a. 4 plots must be assigned: There were $n = 4$ with $\tilde{\sigma} = \frac{1}{n-1}$, $n = 4$ with $\tilde{\sigma} = \frac{1}{n}$, $n = 25$ with $\tilde{\sigma} = \frac{1}{n-1}$ and $n = 25$ with $\tilde{\sigma} = \frac{1}{n}$.
 - b. Show $E[l_i] = \frac{\theta}{2}$ after repeated measurements.
 - c. Show $\theta = \frac{1}{n}(l_1 + \dots + l_n)$ is unbiased.
3. Regression. Aubergine is planted and growth is dependent on fertiliser and water.
 - a. Name two conditions that residuals (ϵ) must fulfil.
 - b. Show that the minimised squared error = maximised likelihood. $E(w)$ was given.
 - c. What else do we need to know to calculate the Bayesian Inference?
4. Hypothesis test: Is there a difference in lifespan between spinach eaters and non-spinach eaters. The p-value is 0,01.
 - a. Write down H_0 .
 - b. Statements true or false:
 - „... disprove H_0 “
 - „... reject H_0 “
 - Out of 1000 spinach eaters, 990 are significantly different from non-spinach eaters.
 - There is a difference between spinach eaters and non-spinach eaters.
 - Spinach eaters live longer than non-spinach eaters.
 - c. Table with values was given: μ for spinach eaters, μ for non-spinach eaters, variance in between and some other values. What percentage of people who are 80 years old are spinach eaters?
5. Causality

Probability of people who write an A grade and get scholarships: $\frac{3}{8}$
 Probability of people who get a sports scholarship: 20%

 - a. Show that $P(\text{get a scholarship}) = 50\%$.
 - b. Show that 75% of the students write an A grade and 40% are good in sport.
 - c. Calculate $P(\text{good in sport, write an A grade} \mid \text{get a scholarship})$.
 - d. Draw the dependence of scholarship, A grade and sport. Explain why it is not possible to infer the population.
6. PCR
 - a. Describe PCR in words. Name two applications.
 - b. Draw PC1 and PC2 in a plot.
 - c. Name 3 approaches to find out how many PCs are needed.
 - d. Name two approaches that can be taken after a PCR to examine the data more closely.